



Supplementary Material for

Developmental and oncogenic programs in H3K27M gliomas dissected by single-cell RNA-seq

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Other Supplementary Material for this manuscript includes the following:
(available at www.sciencemag.org/content/360/6386/331/suppl/DC1)

Tables S1 to S8 as separate Excel files

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SUPPLEMENTARY MATERIALS

Materials and Methods

Tumor acquisition and preparation

Patients and their parents at Boston Children's hospital and the Medical University of Vienna were consented preoperatively in all cases according to Institutional Review Boards. Fresh tumors were collected at the time of surgery and presence of malignant cells was confirmed by frozen section. Tumor tissues were mechanically and enzymatically dissociated using a papain-based brain tumor dissociation kit (Miltenyi Biotec) as previously reported (5, 6).

Fluorescence-activated cell sorting (FACS)

Tumor cells were blocked in 1% bovine serum albumin in Phosphate Buffered Saline (BSA / PBS). Cells were washed with cold PBS, and then resuspended in 1.5 mL of BSA / PBS containing 1uM calcein AM (Life Technologies) and 0.33uM TO-PRO-3 iodide (Life Technologies) to co-stain for 15 min before sorting. Sorting was performed with FACS Aria Fusion Special Order System (Becton Dickinson) using 488nm (calcein AM, 530/30 filter) and 640nm (TO-PRO-3, 670/14 filter) lasers. Non-stained controls were included with all tumors. Standard, strict forward scatter height versus area criteria were used to discriminate doublets and gate only singleton cells. Viable cells were identified by staining positive with calcein AM but negative for TO-PRO-3. Single cells were sorted into 96-well plates containing cold TCL buffer (Qiagen) containing 1% beta-mercaptoethanol, snap frozen on dry ice, and then stored at -80°C prior to whole transcriptome amplification, library preparation and sequencing.

Single cell RNA-seq data generation and processing

Whole transcriptome amplification, library construction and sequencing were performed as previously published (5, 6). Expression levels were quantified as $E_{i,j} = \log_2(\text{TPM}_{i,j}/10+1)$, where $\text{TPM}_{i,j}$ refers to transcript-per-million for gene i in sample j , as calculated by RSEM (16). TPM values were divided by 10 since we estimate the complexity of single cell libraries in the order of 100,000 transcripts and would like to avoid counting each transcript ~ 10 times, as would be the case with TPM, which may inflate the difference between the expression level of a gene in cells in which the gene is detected and those in which it is not detected. For each cell, we calculated two quality measures: the number of genes for which at least one read was mapped, and the average expression level of a curated list of housekeeping genes. We then conservatively excluded all cells with either fewer than 2,500 detected genes or an average housekeeping expression level (E , as defined above) below 2.5. For the remaining cells, we calculated the aggregate expression of each gene as $E_a(i) = \log_2(\text{average}(\text{TPM}_{i,1\dots n})+1)$, and excluded genes with $E_a < 4$. For the remaining cells and genes, we defined relative expression by centering the expression levels, $E_{r,i,j} = E_{i,j} - \text{average}[E_{i,1\dots n}]$. On average, we detected $\sim 5,300$ genes per cell.

Correlation analysis of single-cell expression profiles

Pearson correlation coefficients between expression profiles of all cells that passed quality filtering was calculated using centered gene expression levels. Cells were ordered by hierarchical clustering using 1-correlation coefficient as the distance and complete linkage across all cells (**fig. S3A**), or within each sample and the presumed normal cell types (**Fig. 1A**). The same distance was used for tSNE dimensionality reduction (**fig. S3C,D, S6B**). We used the Rtsne implementation in R (version 3.3.0) with the following non-default parameters: `pca=F`, `is_distance=T`.

Definition of single-cell gene signature scores

Given a set of genes (G_j) reflecting an expression signature of a specific cell type or biological function, we calculate for each cell i , a score, $SC_j(i)$, quantifying the relative expression of G_j in cell i , as the average relative expression (Er) of the genes in G_j , compared to the average relative expression of a control gene-set (G_j^{cont}): $SC_j(i) = \text{average}[Er(G_j, i)] - \text{average}[Er(G_j^{cont}, i)]$. The control gene-set is defined by first binning all analyzed genes into 25 bins of aggregate expression levels and then, for each gene in the considered gene-set, randomly selecting 100 genes from the same expression bin. In this way, the control gene-set has a comparable distribution of expression levels to that of the considered gene-set, and the control gene set is 100-fold larger, such that its average expression is analogous to averaging over 100 randomly-selected gene-sets of the same size as the considered gene-set.

Whole exome and whole genome sequencing

DNA and RNA was extracted from single core biopsies using the AllPrep DNA/RNA extraction kit (Qiagen). For whole genome sequencing (WGS), DNA was randomly fragmented, and libraries (Kapa Biosciences) prepared for paired-end sequencing (2x150bp reads) on an Illumina HiSeq 2500. DNA from germline control was sequenced to 30X coverage and DNA from tumors to 60X coverage.

For whole exome sequencing (WES), fragmented DNA was subjected to library preparation for whole-exome sequencing using the Illumina exome as previously described (17, 18). Flowcell cluster amplification and sequencing were performed according to the manufacturer's protocols using either the HiSeq 2000 v3 or HiSeq 2500. Each run was a 76bp paired-end with a dual eight-base index barcode read. Tumor DNA was sequenced to 150X coverage.

Output from Illumina software was processed by the Picard processing pipeline to yield BAM files containing aligned reads to the NCBI Human Reference Genome Build hg19 with well-calibrated quality scores (19, 20). Sample contamination by DNA originating from a different individual was assessed using ContEst (21). Somatic single nucleotide variations (sSNVs) were then detected using MuTect (22). Following this standard procedure, we filter sSNVs by (1) removing potential DNA oxidation artifacts (23); (2) removing events seen in sequencing data of a large panel of ~8,000 TCGA normal samples; (3) realigning identified sSNVs with NovoAlign (www.novocraft.com) for WES and with BLAT for WGS (24), and performing an additional iteration of MuTect with the newly aligned BAM files. sSNVs were finally annotated using Oncotator (25). Copy-ratio profiles were inferred using ReCAPSEG (<http://gatkforums.broadinstitute.org/gatk/categories/recapseg-documentation>). Read depth at capture probes in tumor samples was normalized using a panel of normal samples to model noise and other biases.

Analysis of point mutations in single-cell data

To detect gene mutations in the single-cell expression data (**Fig. 1D**, **fig. S7**, **S8**), sequencing reads were first aligned to the human genome NCBI Human Reference Genome Build hg38 using STAR version 2.5.1b. Gene annotations were supplied to guide alignment. Mutations were then quantified in each cell at the genomic position in which they were detected in the WGS/WES data using the htlib pileup engine as implemented in pysam. To quantify mutant and wild-type alleles we required at least two concordant reads per cell for *H3F3A* and *PDGFRA*. For all other genes that are less highly expressed we only required a single read. We then filtered mutations that were not detected in any cell, or that were considered erroneously called as they were detected at comparable frequency in samples in which they were not detected in the genome sequencing data compared to samples in which they were detected. The point mutation false positive rate was estimated to be 1.5% and was calculated as the fraction of mutations called in samples in which they were not expected or in the presumed non-malignant cells.

Analysis of CNVs in single-cell data

CNVs were estimated by sorting the analyzed genes by their chromosomal location and applying a moving average to the relative expression values, with a sliding window of 100 genes within each chromosome (**fig. S5**), as we have previously described (4, 5). To check for congruence between gene expression and copy-number alterations in the definition of malignant and non-malignant cells, we scored each cell for the extent of CNV signal, defined as the sum of squares of CNV values across the genome, and for the correlation between the CNV profile of each cell with the average CNV profile of all cells from the corresponding tumor that are classified by expression as malignant (**fig. S5C**). We defined non-malignant cells as those with CNV signal below 0.03 and CNV pattern correlation below 0.2. Six cells for which expression- and CNV-based definitions did not agree were excluded from subsequent analyses and likely represent less frequent normal cell types.

For two tumors (BCH836 and BCH869), we noticed variability within the CNV profiles of the presumed malignant cells, originating from multiple genetic subclones in these tumors. To test our ability to detect different subclones by CNV profiles, we mixed *in silico* two of the tumors (MUV5 and BCH1126) that each had a clonal CNV landscape but were distinct from each other (gain of Chr1q in BCH1126). We simulated 1%, 2%, 5% and 10% of BCH1126 cells mixed with MUV5 cells, in a total of 100, 200, 300, and 400 cells, in 100 iterations each. We performed unsupervised hierarchical clustering of randomly selected CNV profiles (Euclidean distance weighted by the standard deviation of CNV values at each genomic position, average linkage), and cut the resulting dendrogram into two groups (major and minor subclone). If the size of one of the subclones was less than half the number of selected BCH1126 cells, we continued to cut the dendrogram to prevent defining rare outlier cells as subclones. We then assessed our sensitivity and precision at detecting BCH1126 cells in the minor subclone as a function of the total number of cells

profiled (**fig. S6**). Our method for definition of subclones by both CNV profiles and haplotype frequencies is described below.

Analysis of H3F3A mutations using specific primers

To increase the detection rate of H3K27M in single-cell data, we designed primers specific for H3F3A to be included during whole transcriptome amplification (WTA, **fig. S8A**).

Forward primer:

5'-GTCTCGTGGGCTCGGAGATGTGTATAAGAGACAGGAAGCAACTGGCTACAAAAGC

Reverse primer:

5'-TCGTCGGCAGCGTCAGATGTGTATAAGAGACAGGGAAGTTTGCGAATCAGAAGTTC

Both primers were added during the WTA step (**fig. S8A**) at 100nM, while all other steps of the SMART-Seq2 protocol remained unchanged. Primers mimic the integration of adapters by the Tn5 transposase during the Nextera reaction, but in a non-random manner that guarantees coverage of the K27M mutation in resulting sequencing reads. This increased our K27M detection rate in presumed malignant cells from an average of 37% (profiles generated without addition of H3F3A-specific primers) to 97% (profiles generated with addition of H3F3A-specific primers) (**fig. S8D**). We did not detect any apparent change in the expression levels of genes not targeted by the H3F3A-specific primers, and cells clustered with other cells from the same sample that were profiled without the addition of the H3F3A-specific primers (**fig. S8B**).

Definition of malignant and non-malignant gene signatures

Two subpopulations of cells from the MUV5 tumor that are enriched for oligodendrocytic (n=28 cells) and astrocytic (n=88 cells) genes were analyzed in more detail. To verify their definition as malignant cells, we

used definite non-malignant and malignant cells of the respective differentiation lineage from other tumors (as defined in **Fig. 2**) to generate four separate gene signatures that were then applied to the MUV5 subpopulations (**fig. S4**). Gene signatures were generated by applying a permutation test as implemented in the coin package (independence_test function, R version 3.3.0). A P-value threshold of <0.001 was used. To refine signatures, we filtered for genes that were detected in more than 80% of non-malignant oligodendrocytes and less than 20% of OC-like cancer cells, and vice versa. For non-malignant astrocytes and AC-like cancer cells, we used detection thresholds of 70% and 20%.

Identification of differentially expressed genes

When comparing average expression values across tumors (**fig. S9**) we used Analysis of Variance (ANOVA) with an FDR of <0.001, and Tukey's *post-hoc* test (P-value <0.05) to compare between tumor types as indicated. A complete list of differentially expressed genes is given in **table S3**.

When comparing the values of individual cells across tumors and models (**Fig. 4, fig. S19**), we used (1) a two-fold threshold for the average expression; (2) $P < 10^{-4}$ based on a permutation test: we shuffled the assignments of cells to populations 100,000 times and counted the fraction of times where an equal or larger difference was obtained between the two populations; (3) since the different samples vary in the proportions of cells of the same kind of subpopulations (*e.g.*, cycling or differentiated subpopulations), many genes will appear as differentially expressed between samples, although there are no changes in their expression *within* the same subpopulation across the two samples; we wanted to focus on differentially expressed genes which are not driven by the proportions of small subpopulations but instead are expressed by a large proportion of cells in one sample more highly than the highest-expressing cells in the other sample; we thus added a third criterion that at least 30% of the cells in one sample express the gene more highly than the 5% highest expressing cells of the other sample; the rationale for not further increasing this threshold beyond 30% is that genes are often “dropped-out” (*i.e.*, not detected despite being expressed) by a subset of cells in each

sample and hence a higher threshold would exclude many genes. This analysis was repeated for each pairwise comparison of sample types, and genes are included in **Fig. 4D** and **table S8** in each of the comparisons for which they were significant such that the same gene may appear twice; the total number of differentially expressed genes (across the four comparisons) was 641, corresponding to 519 unique genes.

Identification of intra-tumor variability programs using NNMF

For each of the six tumors, Non-Negative Matrix Factorization (as implemented by the Matlab `nnmf` function, with number of factors set to 10) was applied to the centered expression data of malignant cells from that tumor, after converting all negative values to zero. For each of the resulting 10 factors, we considered the 30 genes with highest NNMF scores as a signature for scoring of a variable expression program in that tumor (see “*Definition of single-cell gene signature scores*”). We then aggregated the 60 signatures across the 6 tumors. All malignant cells (across tumors) were then scored for each of the 60 programs. The NNMF programs and the malignant cells were then each ordered by hierarchical clustering (**Fig. 2A, fig. S11A**), revealing three correlated sets of programs (P1, P2 and P3) corresponding to cell cycle and two differentiation programs. For each of those sets, we ranked genes by their correlation with the average cell scores of the programs in the set, and used the top 30 genes to redefine the scores for the three meta-programs.

We then searched for genes preferentially expressed in the remaining cells that do not score highly for any of these three meta-programs. For each of the three meta-programs, we calculated average fold-change and permutation p-value between all cells that score for that program (score>1) and the fourth subpopulation of cells that do not score for any of the programs (score<0.5) and defined differentially expressed genes with a fold-change>2 and FDR<0.05 (**fig. S11B**). No genes were significantly higher in the fourth subpopulation than in all three subpopulations; 20 genes were significantly higher in the fourth subpopulation than in two of the three subpopulations, and, of these, 19 were significantly higher in the fourth subpopulation than in

the AC-like and OC-like cells, but not the cycling cells; these 19 genes were defined as P4 (**Fig. 2B**), and denoted as the OPC-like program as it was preferentially expressed by OPCs (**fig. S11D-F**).

Principal Component Analysis (PCA) of this fourth subpopulation (cells with high expression of the OPC-like program and low expression of the AC-like and OC-like programs) revealed an additional expression program that varied among those cells in each of the tumors (**fig. S11C, table S5**). This program was also highly expressed by OPCs (**fig. S11D-F**). We thus identified two OPC-related programs, one that is expressed across all cells in the fourth subpopulation (P4 in **Fig. 2B**, also denoted as OPC-like) and another that varies between the cells *within* this subpopulation (denoted as OPC-variable, **Fig. 4B**). Taken together, this strongly suggests that the majority of H3K27M-glioma cells reflect an OPC-like state.

Identification of intra-tumor variability programs using hierarchical clustering

As a second approach to identify intra-tumor variability programs we relied on expression signatures associated with malignant cell cluster. First, we used average linkage hierarchical clustering of the single malignant cells from each of the tumors separately, using one minus the Pearson correlation (across all analyzed genes) as the distance metric. In order to select clusters without a pre-defined strict threshold on the number of clusters or their level in the hierarchical tree, we first recovered all sub-trees, and then exclude potential clusters by size, by their signal for differential expression, and by their redundancy with other clusters, in the following ways: (1) We exclude clusters that consist of less than 5 cells or more than 80% of the malignant cells in the respective tumor. (2) For each cluster, we estimate the number of preferentially expressed genes: we identify all genes with 2-fold higher average expression in the cluster than in all other malignant cells from the same tumor, and calculate their p-value using a t-test; we then count the number of significant genes (N_{sig1}) with a Benjamini-Hochberg False Discovery Rate (FDR) below 0.05. We also compare the nominal p-values of genes across different clusters and count the number of genes which are not only significant in each cluster but are more significant in that cluster than in all other clusters from the

same tumor (N_{sig2}). All clusters with either $N_{sig1} > 50$ or $N_{sig2} > 10$ are defined as having sufficient signal of differential expression and are retained for further analysis. (3) For each pair of clusters, we calculate what fraction of cells in the union of the two clusters are in the intersection of the two clusters. If this fraction is above 75%, we exclude the cluster with lower N_{sig2} . Applying this approach to the six tumors revealed 178 clusters, including (as desired) many cases of both a large cluster and its smaller sub-clusters (e.g., 59 clusters in **fig. S12A**). Finally, we use the differentially expressed genes (N_{sig1}) as each cluster's signature yielding 178 signatures. Almost all signatures had significant overlap with one of the five NNMF meta-programs (**fig. S12B-C**). The few signatures that did not overlap with the meta-programs were associated with high expression of pseudogenes and antisense transcripts and may reflect a technical effect (**fig. S12B**).

Comparison of gene expression between H3K27M-glioma and normal brain cell types.

For normal brain cell profiles, we analyzed both scRNA-seq data produced in our laboratory at MGH/Broad, as well as a recently published dataset from the Quake lab in Stanford (11). Both studies used the SMART-Seq2 protocol and include OPCs, OCs, ACs and neurons. Dataset collected in our hands included fetal NPCs (5) and OPCs and OCs newly profiled from a patient with oligodendroglioma; these cells did not contain genetic alterations found in the malignant cells of that patient, expressed markers of the respective cell types and were similar to the respective cell types from the Quake lab datasets. Altogether, for each of seven scRNA-seq datasets of normal brain cell types, we defined average expression levels (E_a) and compared these to the average expression levels of malignant cells from each of the H3K27M gliomas. We first looked for genes with higher expression in H3K27M-glioma than in all the normal cell types examined; for each gene, we compared its expression levels (E_a) among the six H3K27M-gliomas to that in the normal cell type dataset with the highest expression of that gene. We found 80 genes with at least 2-fold higher expression in H3K27M-glioma and a *t*-test *p*-value below 0.05 after FDR correction (**fig. S14; table S6**). Similar search for genes with lower expression in H3K27M-glioma than in all the normal cell types

examined identified only one gene, FOXP1 (**table S6**). We further analyzed the set of H3K27M-glioma-high genes, but did not identify significant functional enrichments. Since we expected PRC2 targets to be enriched in H3K27M-glioma-high genes, but did not observe such enrichment, we further examined the expression of PRC2 targets (10) across normal cell types. We focused on PRC2 targets which are expressed by H3K27M-glioma cells (defined by $E_a > 5$) and as a control we compared them to all other genes (non-PRC2-targets) expressed by H3K27M-glioma cells, using the same threshold. We found that PRC2 targets are significantly higher (compared to non-targets) in OPCs and significantly lower (compared to non-targets) in OCs and in NPCs ($P < 0.05$, t-test).

Lineage and stemness score of individual malignant cells

Cells were first ordered by their stemness, defined as expression of the OPC-shared program minus the maximal expression of the two differentiation programs, and the differentiated cells were further classified by a lineage score distinguishing the oligodendrocytic from astrocytic lineages (**Fig. 2C**).

Comparison between lineage and stemness programs of H3K27M-gliomas and IDH-mutant gliomas

To enable a direct comparison between H3K27M, IDH-A and IDH-O gliomas we combined the malignant cells from all tumors which were processed by the same protocols, and centered the expression of each gene across all cells in the combined dataset. We then scored all cells for the lineage and stemness programs defined here and in previous studies of IDH-A and IDH-O glioma (the gene-sets of IDH-A and IDH-O (5, 6) were combined to define IDH glioma gene-sets). This approach – scoring all cells for gene-sets of both H3K27M- and IDH-mutant gliomas – enabled us to test the specificity of gene-sets to glioma types; we found that the AC-like gene-sets are comparable and highlight the same cells in each of the glioma types, while OC-like and stem-like gene-sets showed specificity, with H3K27M-gliomas scoring primarily for

H3K27M gene-sets and vice versa (**Fig. 3A,C,E**). To further compare the expression programs between tumor types we defined tumor type-specific gene scores for each program (AC-like, OC-like and stem-like). In each tumor type, we compared the cells that score highly for a given program (score>1) with those that score lowly for that program (score<0), to define average expression differences between those subsets and to focus on differentially expressed genes with fold-change above 4 in at least one of the tumor types. We then defined differentially expressed genes which are common (*i.e.*, with fold-change above 2 in the three tumor types), H3K27M-specific (with fold-change above 4 in H3K27M, and less than 1.5 in both IDH-mutant tumors) and IDH-mutant specific (with fold-change above 3 in both IDH-mutant tumors but less than 1.5 in H3K27M). Finally, we defined the fraction of undifferentiated cells in each tumor as those with score <1 for both the AC-like and OC-like programs of the respective tumor type.

Genetic intra-tumor heterogeneity by haplotype analysis

We developed a computational approach for identifying haplotype frequencies from scRNA-seq data, which when combined with single-cell CNV analysis improves the ability to define genetic subclones and helps the inference of their evolutionary relationships (**fig. S16A**). The approach consists of four major steps: the identification of patient-specific heterozygous SNPs in matching bulk WGS data, the phasing of SNPs into haplotypes, the quantification of haplotype frequencies in the single-cell RNA-seq data, and the identification of the putative phylogeny of tumor subclones.

For the identification of patient-specific heterozygous SNPs we used high-coverage bulk whole-genome sequencing data from both germline (*i.e.*, blood) and tumor DNA from two patients that showed multiple subclonal CNVs (BCH836 and BCH869). For each patient, we first called variants from germline DNA using samtools mpileup (version 1.3, arguments -ug) and bcftools call (version 1.1, arguments -vm). We then excluded sites with quality less than 30 using bcftools filter, and retained only heterozygous SNPs using bcftools view (arguments -v snps -m2 -M2 -g het). Remaining positions were called from both

germline and tumor samples (single biopsy for BCH836, biopsy and six different tumor regions from autopsy tissue for BCH869) using samtools mpileup (arguments -v -t DP,AD --max-depth 1000000). The resulting vcf files were processed in R (version 3.3.0) with the VariantAnnotation Bioconductor package (version 1.18.7). For each patient, positions with less than 0.5 or more than 1.5-fold coverage in the germline sample compared to the `_genome` average were excluded. Next, positions with an allele frequency less than 1/4 or more than 3/4 in the germline sample were excluded. Remaining positions (≥ 2 million) were considered high-confidence heterozygous SNPs and used for subsequent analysis. Most of these SNPs have been previously identified in the human population by projects such as the HapMap project or the 1,000 genomes project.

For phasing of haplotypes, we only considered the SNP allele frequencies in the bulk tumor WGS data that lie within chromosomal or sub-chromosomal regions that showed broad copy-number variations (**fig. S16B,C**). The allele frequency of these SNPs deviates from 0.5 in most of these regions (e.g. is close to 0 or 1 for regions that underwent LOH, 0.33 or 0.67 for a single gain of an allele), allowing for phasing of SNPs into haplotypes. We excluded SNPs that showed an allele frequency between 0.45 and 0.55 in the bulk tumor WGS data, as these could not be confidently phased (e.g. SNPs that lie within genomic regions showing copy-number variations that affect only small subpopulations of cells and do not have a strong impact on the bulk allele frequency). Using this approach, typically $\geq 10,000$ SNPs per genomic region were combined in a haplotype.

To account for the much sparser and more uneven coverage of SNPs in the scRNA-seq data, we jointly quantified all SNPs in each haplotype (and thus are not as susceptible to drop-outs, which should impact each SNP independently): first, the scRNA-Seq reads were aligned to the hg19 reference genome using STAR (version 2.5.2b), guided by gene annotations. Second, allele read counts for all SNPs as defined above were generated using samtools mpileup (arguments -v -t DP,AD --max-depth 1000000). Third, read counts supporting each haplotype were summed up for each single cell (**fig. S16D**). In this way, depending

on the size of the region and the prevalence of expressed heterozygous SNPs in the respective sample, up to thousands of supporting reads were combined, providing accurate estimates of haplotype frequencies at the single-cell level (**Fig 3I, J, fig. S17B**).

Single-cell haplotype frequencies were jointly analyzed with CNV profiles to define tumor subclones and infer their evolutionary relationships (**Fig. 3L, fig. S17D**). Subclones were first defined by hierarchical clustering of a distance matrix derived from both genome-wide single-cell haplotype frequencies and CNV profiles, and then further refined by considering only select chromosomal regions that differ between subclones on a per-sample basis. The use of haplotype frequencies improved our ability to define genetic subclones, and enabled assignment of distinct haplotype and copy-number states to individual subclones (**table S7**). In many instances, this provided an additional layer of information that enabled inference of the putative phylogeny of different subclones. Phylogenetic trees were inferred by ordering subclones by increasing number of alterations (starting from the diploid ground state), assuming that alterations are more likely to occur than be reversed (e.g. BCH869 subclone 3 deriving from subclone 2 by gain of chromosome 2q). Alterations that were incompatible with this notion resulted in branching of the phylogenetic tree. The use of haplotype frequencies provided an additional layer of evidence that guided inference, as it is impossible for haplotypes to be regained after they have been previously lost, ruling out some ancestry relations. For example, the loss of different alleles of chromosome 14 in sample BCH869 indicates that subclone 1 does not derive from subclone 2, but both subclones originate from an earlier, unobserved ancestor. Similarly, as subclone 3 shares the same allele of chromosome 14 with subclone 2, it does not derive from subclone 1, but has independently gained (a different allele of) chromosome arm 2q.

RNA in situ hybridization

Paraffin-embedded tissue sections from human tumors from Boston Children's Hospital, the Medical University of Vienna, Austria, the Medical University of Ljubljana, Slovenia, and Hospital Sant Joan de

Deu Barcelona, Spain, were obtained according to Institutional Review Board-approved protocols. Sections were mounted on glass slides and stored at -80°C. Slides were stained using the RNAscope 2.5 HD Duplex Detection Kit (Advanced Cell Technologies, Cat. No. 322430). Slides were baked for 1 hour at 60°C, deparaffinized and dehydrated with xylene and ethanol. The tissue was pretreated with RNAscope Hydrogen Peroxide (Cat. No. 322335) for 10 minutes at room temperature and RNAscope Target Retrieval Reagent (Cat. No. 322000) for 15 minutes at 98°C. RNAscope Protease Plus (Cat. No. 322331) was then applied to the tissue for 30 minutes at 40°C. Hybridization probes were prepared by diluting the C2 probe (red) 1:50 into the C1 probe (green). Advanced Cell Technologies RNAscope Target Probes used included Hs-MKI67 (Cat. No. 591771-C2; 591771), Hs-PDGFA (Cat. No. 604481-Cs) and Hs-ApoE (Cat. No. 433091). Probes were added to the tissue and hybridized for 2 hours at 40°C. A series of 10 amplification steps were performed using instructions and reagents provided in the RNAscope 2.5 HD Duplex Detection Kit. Tissue was counterstained with Gill's hematoxylin for 25 seconds at room temperature followed by mounting with VectaMount mounting media (Vector Laboratories).

Cell culture

Human H3K27M glioma cell lines (BCH869, BCH245, DIPG012) were derived from patients treated at Boston Children's Hospital, Massachusetts General Hospital and Hospital Sant Joan de Deu Barcelona, Spain according to Institutional Review Board-approved protocols. Human IDH-wildtype glioblastoma lines MGG4 and MGG6 were previously characterized (15). For all functional experiments, H3K27M cell cultures were grown as gliospheres in Tumor Stem Medium (TSM) base supplemented with B27 Minus Vitamin A, EGF, FGF, PDGFA, PDGFB and heparin (26). All GBM cell cultures were grown as gliospheres in Neurobasal Medium, supplemented with N2 supplement, B27 supplement, Glutamax, Pen/Strep (all Life Technologies), EGF and FGF (Shenandoah Biotechnology INC). For differentiated glioma cell (DGC) conditions, H3K27M cells were either (i) cultured in TSM base supplemented with 7.5%

or 10% Fetal Bovine Serum (Atlas Biologicals) in TC-treated flasks or (ii) cultured in gliomasphere TSM base supplemented with B-27 Minus Vitamin A with or without EGF, FGF, PDGFA, PDGFB and heparin in non-TC treated flasks (SF_Adh and GF⁻, respectively) (**Fig. 4A,B, fig. S18,19**). Viability was determined by CellTiter-Glo Luminescent Cell Viability Assay (Promega) at the end of treatment (day 7 or day 10). For drug treatment experiments, PTC-209 (SML1142, Sigma-Aldrich) was suspended in DMSO and used at the concentrations indicated.

Intracranial xenotransplantation

The primary H3K27M-glioma cell line BCH869, grown as gliomasphere in serum-free conditions (“gliomaspheres”) or under adherent conditions in 10% serum as differentiated glioma cells (“DGC 10%”) were injected stereotactically into the right striatum of 5 weeks-old female NSG mice (NOD.Cg-Prkdc^{scid} Il2rg^{tm1Wjl}/SzJ, The Jackson Laboratory, Bar Harbor, ME). Briefly, mice were anesthetized with 2% isoflurane mixed with medical air, and placed on a stereotactic frame. The skull of the mouse was exposed through a small skin incision, and a small burr hole was made using a 25-gauge needle at the selected stereotactic coordinates. The BCH869 cells (100,000 or 200,000 cells in 2 μ L PBS) were loaded on a 33-gauge Hamilton syringe, and injected slowly using the following coordinates: 2.0 mm lateral of the bregma, and 2 mm deep to the cortical surface. Upon completing injection, the needle was left in place for another minute, then withdrawn slowly to help reduce cell reflux. After closing the scalp with suture and staple, mice were returned to their cages placed on a warming pad and visually monitored until full recovery. Mice were then checked daily for signs of distress, including seizures, ataxia, weight loss, and tremors, and euthanized as they developed neurological symptoms, including head tilt, seizures, sudden weight loss, loss of balance, and ataxia. All animal studies were performed according to Dana-Farber Cancer Institute Institutional Animal Care and Use Committee (IACUC)-approved protocols.

Small animal MRI

MRI experiments were performed on a Bruker BioSpec 7T/30 cm USR horizontal bore Superconducting Magnet System (Bruker Corp., Billerica, MA) equipped with the B-GA12S2 gradient and integrated with up to 2nd order room temperature shim system, which provides a maximum gradient amplitude of 440 mT/m and slew rate of 3440 T/m/s. The Bruker-made 23 mm ID birdcage volume radiofrequency (RF) coil was used for both RF excitation and receiving. The Bruker AutoPac with laser positioning was used for accurate definition of the region of interest. Animals were anesthetized with 1.5% isoflurane mixed in medical air at a flow rate of 2 L/min. Body temperature was maintained at 37°C using a warm air fan. A pressure-transducer for respiratory gating was placed on the abdomen. Animal respiration and temperature were monitored and regulated by the SAI (Sa Instruments Inc., Stony Brook, NY) monitoring and gating system model 1025T. Bruker Paravision 6.0.1 was used for MRI data acquisition. T2-weighted images were obtained by a fast spin echo (RARE) sequence with fat suppression using the following parameters: TR = 6000 ms, TE = 36 ms, FOV = 19.2 x 19.2 mm², matrix size = 256x192, spatial resolution = 75 x 100 µm², slice thickness = 0.5 mm, number of slices = 29, rare factor = 15, number of averages = 8, acquisition time 7 min. Images were analyzed and tumor volumes extracted using the semi-automatic segmentation analysis software ClinicalVolumes (ClinicalVolumes, London, UK). Osirix imaging software was used to generate 3D reconstructed images.

Cas9-mediated gene knockout

CrisprRNA and tracerRNA were synthesized by Integrated DNA Technologies (Coralville, IA) and co-incubated to form guide RNA (gRNA) complexes (95°C for 5 minutes) based on the manufacturer's instructions). gRNA complexes were incubated with Cas9 protein (provided by QB3 Macrolab, University of California Berkeley, Berkeley, CA) at 37°C for 15 minutes to assemble ribonucleoprotein (RNP) complexes. Next, RNP complexes (3 µL) were electroporated together with corresponding single-stranded

homology-directed repair (HDR) oligonucleotides (200 pmol) into the indicated cells lines using a 4D-Nucleofector 96-well shuttle system (Lonza Group LTD, Basel, Switzerland) with program DS-150. Genomic DNA was extracted from electroporated cells at the indicated time points and the relevant DNA segments were amplified by polymerase chain reaction (PCR) to assess the efficiency of editing. The PCR products were purified from 1% agarose gels using MinElute Gel Extraction columns (Qiagen, Hilden, Germany). Purified DNA was analyzed by Sanger sequencing (Quintarabio, South San Francisco, CA). The efficiency of Cas9-mediated gene editing was determined by comparing DNA sequences from cells electroporated with control RNPs (Cas9 and non-targeting lacZ gRNA) or target RNPs (Cas9 + gene-specific gRNA) using Tracking of Indels by Decomposition (TIDE) analysis (27).

CrisprRNA sequences and Ultramer Repair Templates

CrisprRNA (crRNA) were 20 nucleotides in length and were obtained from Integrated DNA Technologies with the following sequences: BMI1-3: ACCACTACTGAATATAAGGT; BMI1-5: CAAAGCACACACATCAGGTG; PDGFRA-1: TGTGACTTTCGCCAAAGTGG; PDGFRA-5: GTAACCTTACACAACAGTGA. Homology directed repair (HDR) templates were obtained from Integrated DNA Technologies as single stranded ultramers that were 192 nucleotides in length with the following sequences specific for the following crRNA cut sites:

BMI1-3 HDR:

TAGTCTGTAAAACGTGTATTGTTTCGTTACCTGGAGACCAGCAAGTATTGTCCTATTTGTGATG
TCCAAGTTCACAAGACCAGACCACTACTAGTTAGCTAACAGGAACTGTTGAAATTCCTTGT
TTGTAATTATTATTGGAGTTGTATAATTTACTGAAGGCAACCCTCTTTATTTCTTCACAGAAA
ATTT;

BMI1-5 HDR:

ATTATGGCCATTATTTCTGTGTCTTGCAGGATTTTTTATCAAGCAGAAATGCATCGAACAACG
AGAATCAAGATCACTGAGCTAAATCCCTAAACCGGTTAAGTGCTTTGTGGAGGGTACTTCAT
TGATGCCACAACCATAATAGAATGTCTACATTCCTGTAAGTACCGAGCTTTAGCTCTCTTTTG
TATC;

PDGFRA-1 HDR:

CTTCCTGGACTATTTTGGCCAACAATGTCTCAAACATCATCACGGAGATCCACTCCCGAGAC
AGGAGTACCGTGGAGGGCCGTGTGACTTTAGATAGCTAACAGGAGACCATCGCCGTGCGAT
GCCTGGCTAAGAATCTCCTTGGAGCTGAGAACCGAGAGCTGAAGCTGGTGGCTCCCAGTGA
GTTCTCA;

PDGFRA-5 HDR:

AGGAATGACGGATTATTTAGTCATCGTGGAGGATGATGATTCTGCCATTATACCTTGTCGCA
CAACTGATCCCGAGACTCCTGTAACTTTAGTTAGCTAACGGGGGTGGTACCTGCCTCCTAC
GACAGCAGACAGGGCTTTAATGGGACCTTCACTGTAGGGCCCTATATCTGTGAGGCCACCGT
CAAAGG.

The following oligonucleotides were used to amplify the edited DNA segment for Sanger sequencing and TIDE analysis.

BMI1 FWD: GGGGATTGTGTGGCGTCTG; BMI1 REV: TAGGGAATAAAGAGGAATGGAAGCC;
BMI1-3 Sequencing: TGTTGGTACAAAGTGGTGAAG; BMI1-5 Sequencing:
AGTTTGGTAGAACTGATTCCG.

PDGFRA-1 FWD: CTGAGGATCATCGCAACCCT; PDGFRA-1 REV:
GGGCAGACACCTCTACTTCAT; PDGFRA-1 Sequencing: AGACAAGGTCCCAACTCCTTGCCAT.
PDGFRA-5 FWD: TCCATCAGGAGACAGGCAAT; PDGFRA-5 REV:
TAGGGCCGCCATAGTCAGGA; PDGFRA-5 Sequencing: TCAATAATGCCAGTGGGATAG.

Supplemental Figures

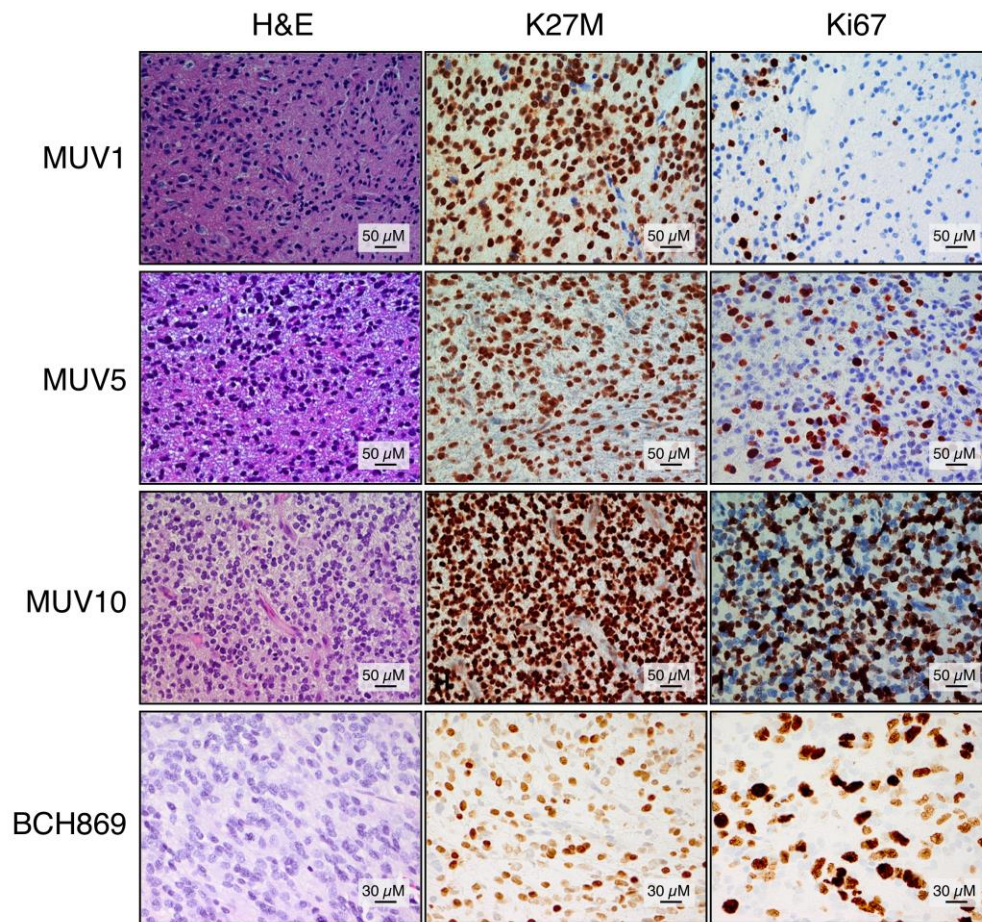


Figure S1. Representative histology of four H3K27M gliomas in the cohort. Hematoxyline and eosin (H&E) stains of tumors shows densely cellular tumors with important cytonuclear pleomorphism. H3K27M immunohistochemistry shows strong nuclear immunoreactivity of malignant cells. Ki-67 immunohistochemistry is used for proliferation and shows estimated rates of 10%, 20%, 60% and 25% of cycling cells for tumors MUV1, MUV5, MUV10, and BCH869 respectively.

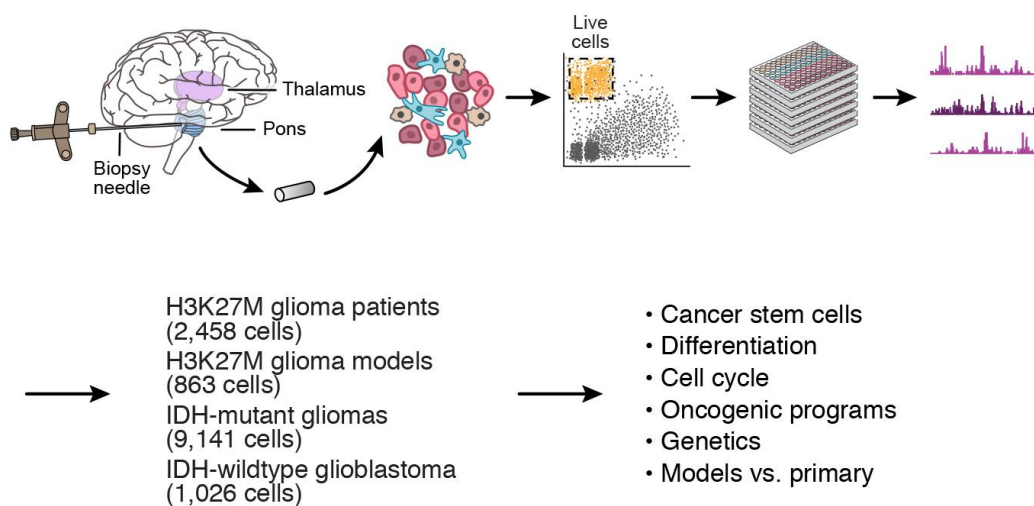


Figure S2. Workflow. Samples are collected at the time of surgery, freshly dissociated to single-cell suspensions and sorted based on viability markers into 96-well plates. Processing to full-length scRNA-seq is performed following the Smart-Seq2 protocol. Data from H3K27M patient samples and models are analyzed and compared to other classes of gliomas profiled by scRNA-seq.

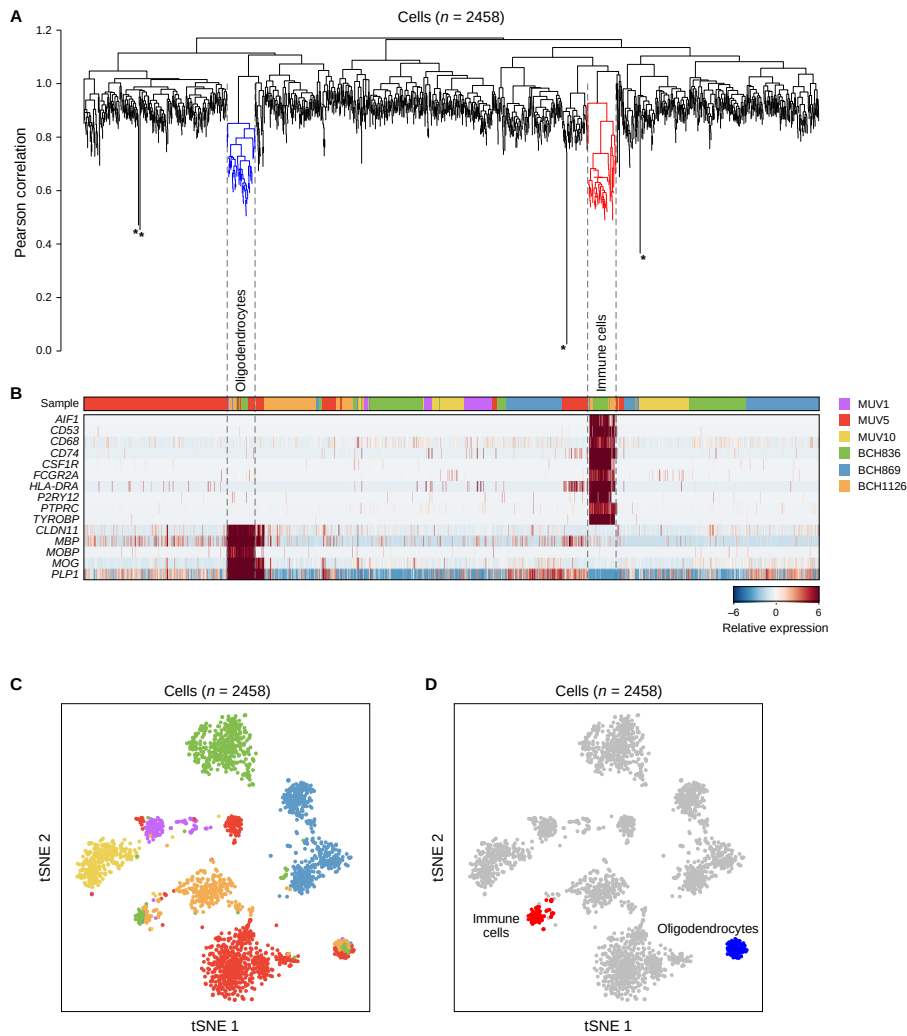


Figure S3. Unsupervised clustering of H3K27M-glioma single-cell expression data. (A) Hierarchical clustering using 1-Pearson correlation as distance measure and complete linkage of 2,458 cells that pass quality filtering. Two clusters of cells show particularly high cell-to-cell correlations and express markers of non-malignant cell populations (see B). Four pairs of cells showed very high correlation (indicated by*), and might reflect a technical artefact, as most of them were located on neighboring wells after flow sorting. These cells were removed from subsequent analysis. (B) Heatmap of relative gene expression levels for genes previously described as markers for microglia and oligodendrocytes. (C) 2D representation of sample correlations by tSNE dimensionality reduction, color-coded by tumor sample as in (B). (D) Identical 2D representation, color-coded for non-malignant cell populations as defined in (A, B). (see also Fig. 1 and fig. S4-S7 for a more detailed analysis).

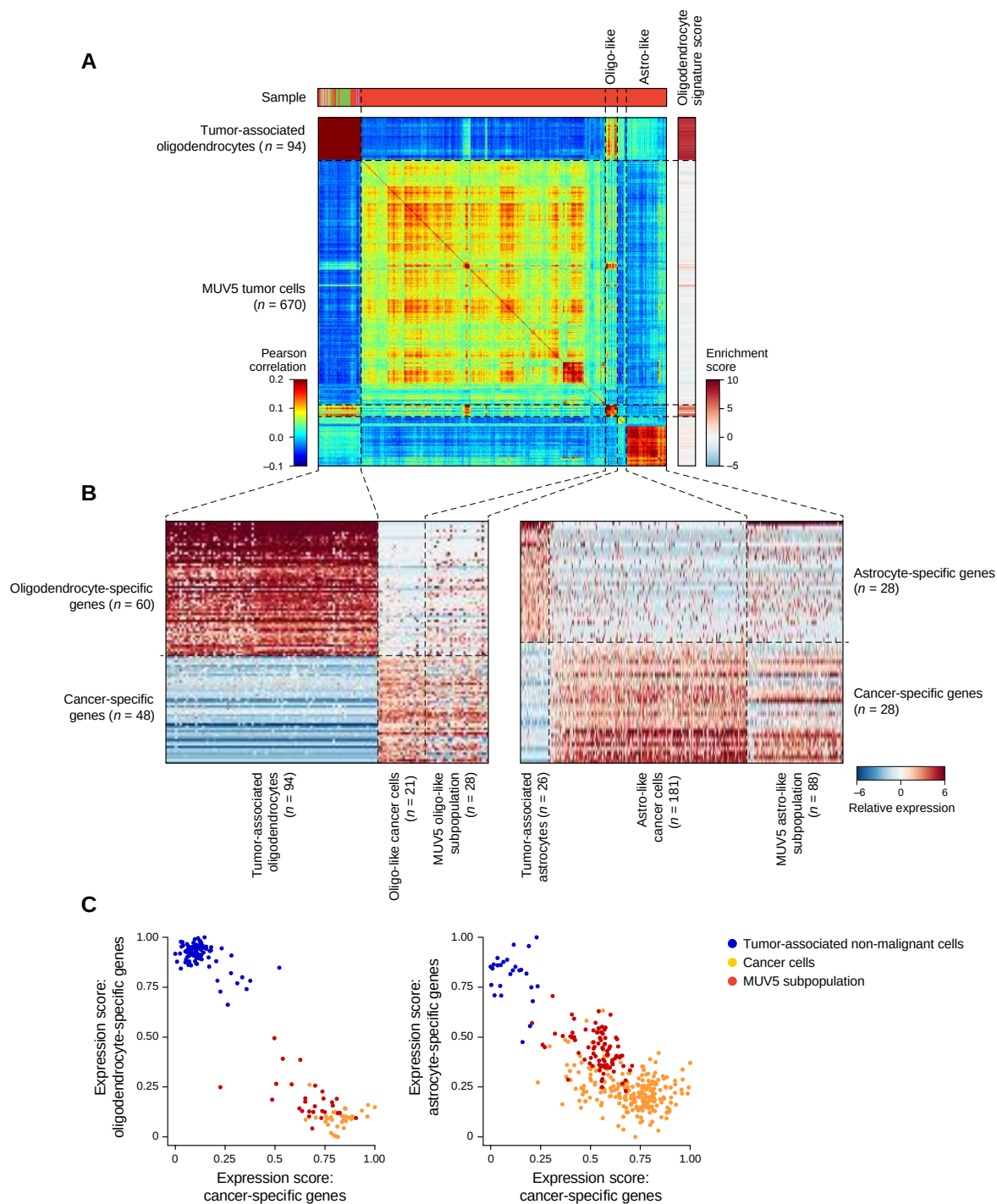


Figure S4. Detailed analysis of MUV5 cells and non-malignant cells in our cohort. (A) Heatmap representation of cell-to-cell correlations as presented in Fig. 1A, restricted to cells from sample MUV5 and non-malignant oligodendrocytes. MUV5 tumor cells ($n=670$) form a cluster of 28 cells enriched for oligodendrocytic genes (OC-like cells), and another cluster of 88 cells enriched for astrocytic genes (AC-like cells). **(B)** Heatmap representation of relative gene expression levels for genes defined as either specific

to OC-like cancer cells (from definitive malignant cells with detected genetic alterations), or specific to non-malignant oligodendrocytes, showing that MUV5 OC-like cells are more similar to malignant cells from other tumors than to non-malignant oligodendrocytes (left). Similar heatmap representation for genes specific to AC-like cancer cells (from definitive malignant cells with detected genetic alterations), or non-malignant astrocytes, supporting MUV5 AC-like cells are more similar to malignant cells from other tumors than to non-malignant astrocytes (right). (C) Signature scores for gene sets as defined in (B).

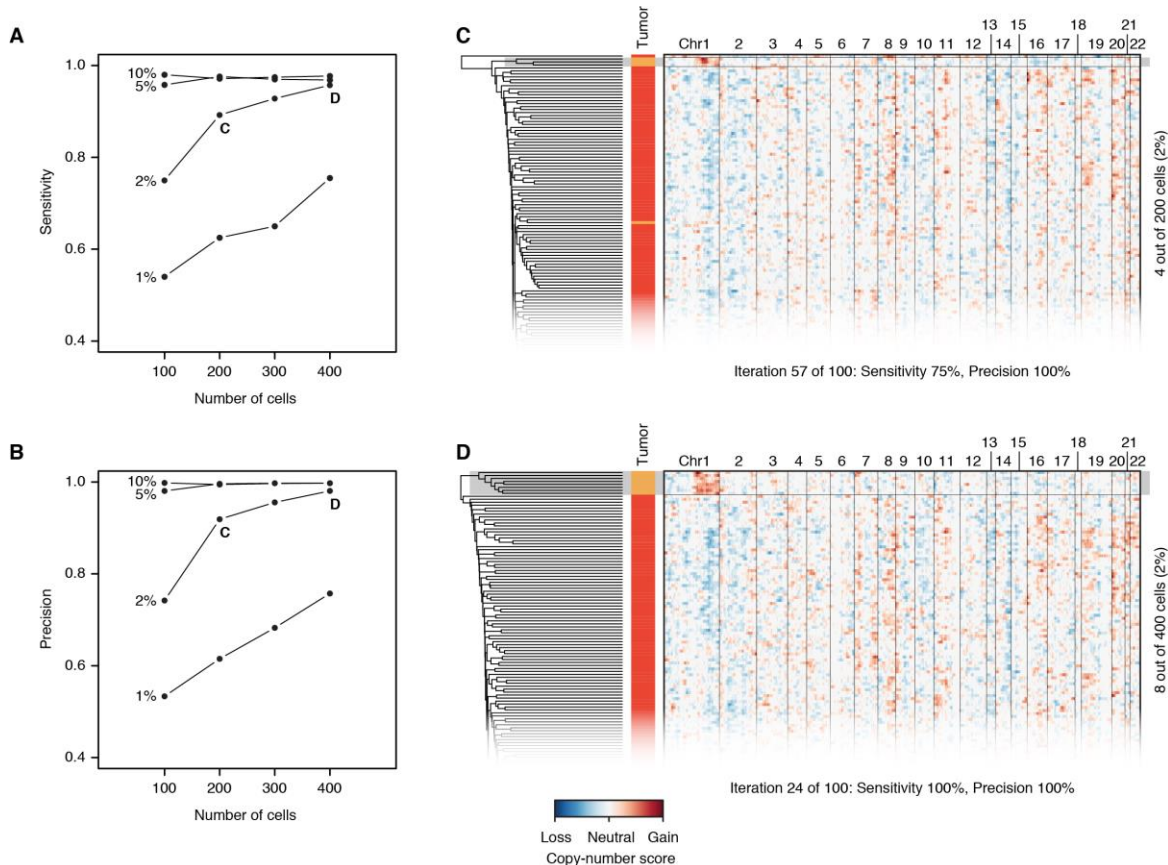


Figure S6: *In silico* simulation to assess sensitivity and precision of CNV inference. (A) Sensitivity (Y axis) in detecting BCH1126 cells mixed at 1%, 2%, 5% and 10% with MUV5 as a function of the number of cells profiled (X axis) (8). The average sensitivity of 100 iterations is shown. C and D labels indicate the examples shown in panels C and D. (B) Precision (Y axis, fraction of cells from BCH1126 in minor subclone) per number of cells (X axis) for the same *in silico* experiment as in (A). For a clone present at 5%, high sensitivity and precision was achieved when a total of 100 cells are profiled. For a clone present at 2%, high sensitivity and high precision was achieved when profiling at least 400 cells. (C) Example of CNV analysis for 2% of BCH1126 (orange) mixed with 98% MUV5 (red), 200 cells profiled. The automatically detected minor subclone is indicated by a grey box. (D) Example of CNV analysis for 2% of BCH1126 (orange) mixed with 98% MUV5 (red), 400 cells profiled.

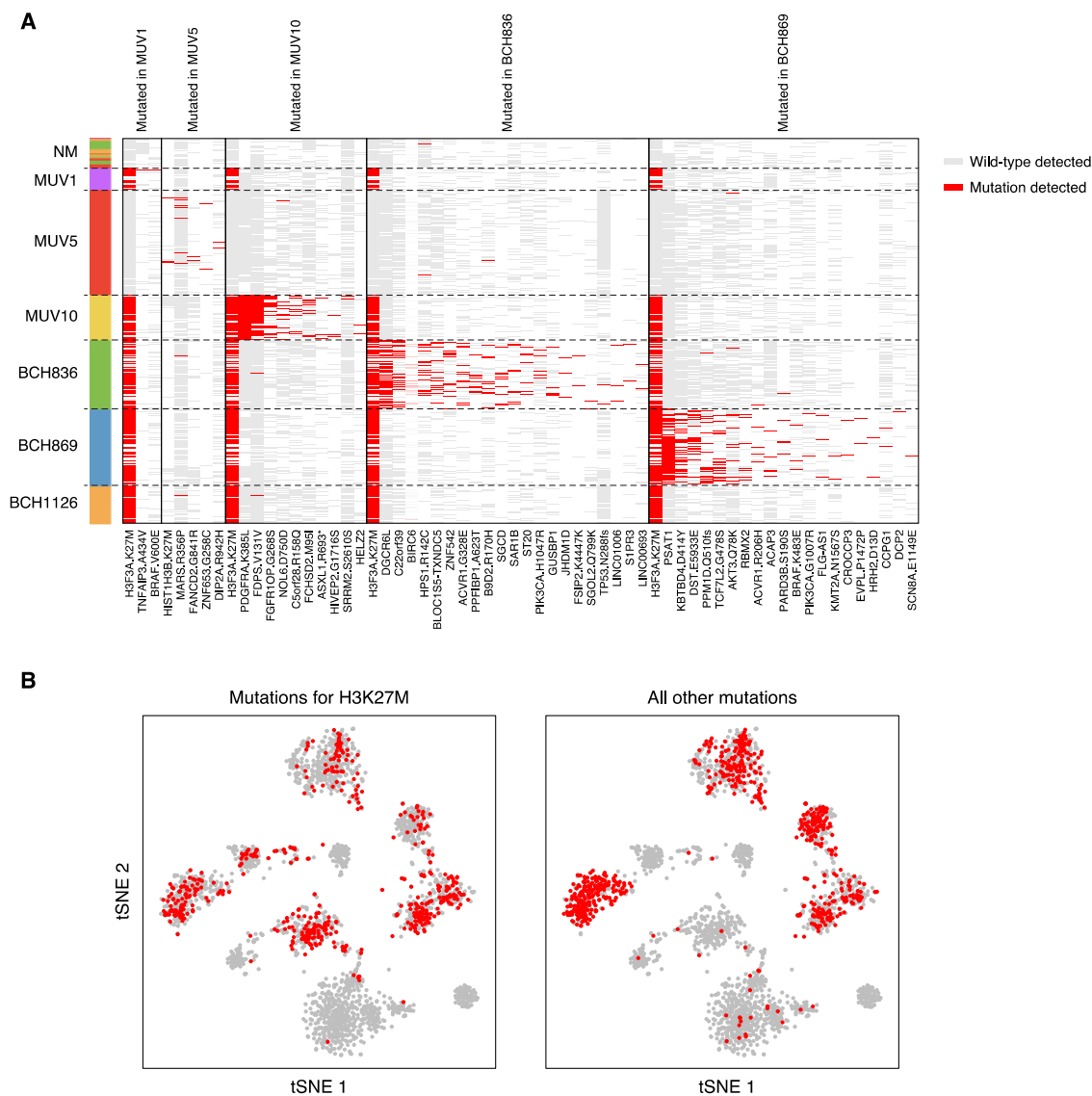


Figure S7. Detection of genetic alterations in single-cell data. (A) Evidence for mutations in genes identified as mutated by WGS/WES in scRNA-seq data. Genes that showed evidence for a mutation in one (MUV5) or at least two (all other tumors) cells are included. The capacity to detect a specific event depends on expression levels of a given gene and is limited by the sensitivity of scRNA-seq protocols. In addition, note that in one of the six tumors (MUV5), the H3K27M mutation is in histone H3.1 (*HISTH3B/C*). Since H3.1 transcripts are not polyadenylated (unlike H3.3), they are not effectively captured by the SMART-Seq2 protocol and consequently the mutation was only detected in six cells in that tumor. **(B)** tSNE map as presented in **fig. S3B**, color-coded by gene mutations identified in *H3F3A* or *HIST1H3B* (red, left), or all other mutations (red, right).

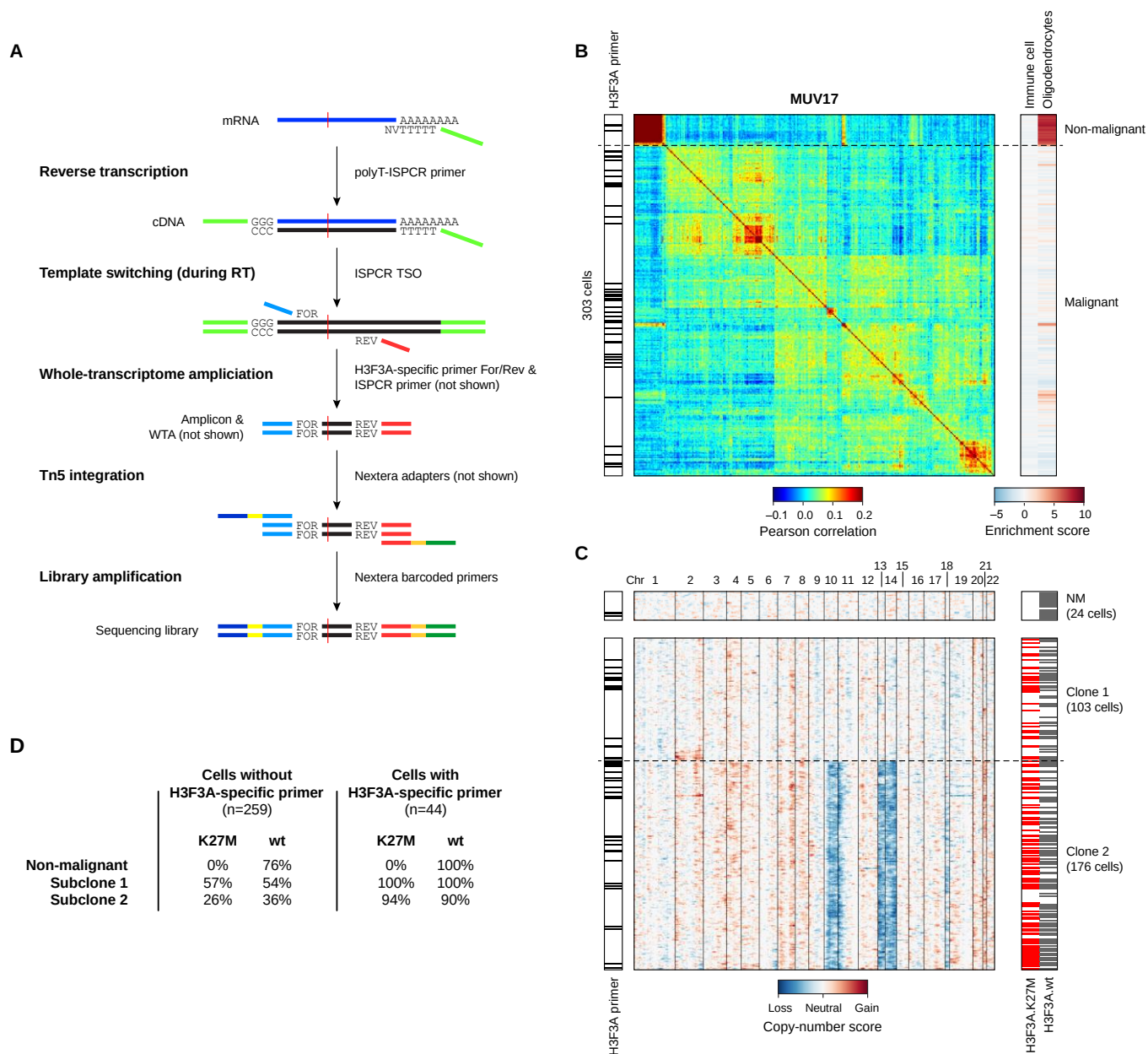


Figure S8. Inclusion of H3F3A-specific primers during WTA increases detection of H3K27M in SMART-Seq2 scRNA-Seq. (A) Modified workflow of SMART-Seq2 protocol including the addition of H3F3A-specific primers at the whole-transcriptome amplification (WTA) step. The primers introduce identical adapter sequences as in the Nextera step, but at specific positions in the transcript that will allow for the H3K27M mutation to be covered in the sequencing reads (8). (B) Pearson correlation of scRNA-Seq profiles (colorbar) of non-malignant and malignant cells (marked on right) from an additional tumor

sample (MUV17). Cells in the non-malignant cluster have high enrichment score for oligodendrocyte signature genes (right), and also co-cluster with all other non-malignant cells derived from patient samples in this study. Left panel notes whether H3F3A-specific primers were included in each cell library (black) or not (white). **(C)** Single-cell CNV profiles inferred for each cell (rows) from MUV17 (chromosomes in columns) along with the presence of H3K27M mutation (red) and WT (grey) alleles. Top matrix: non-malignant cells. Bottom matrix: malignant cells. Dashed line: two malignant subclones, one of which does not show clear copy- number alterations. **(D)** Percentages of MUV17 cells with H3K27M or H3F3A wildtype detection, with and without the inclusion of H3F3A-specific primers during WTA.

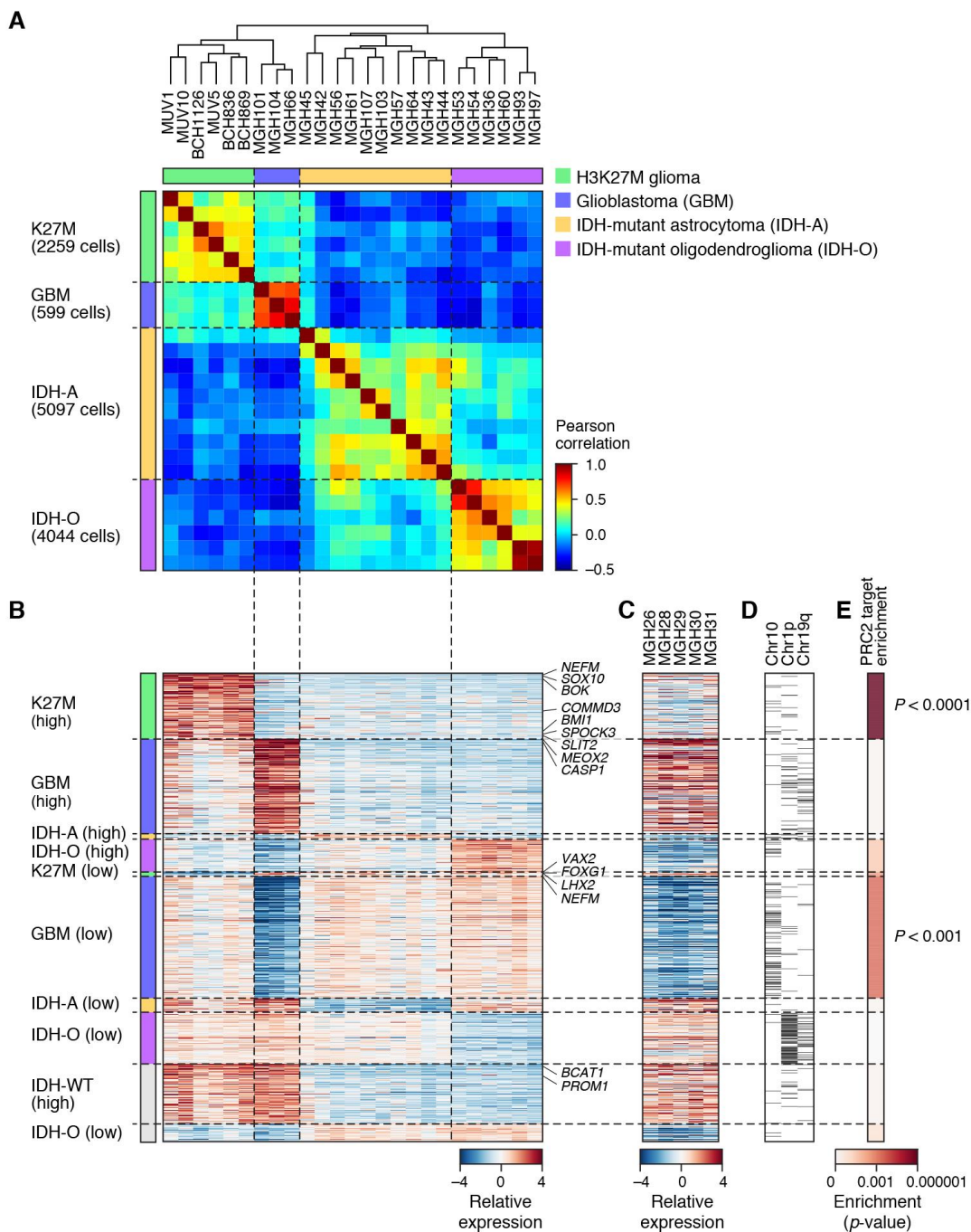


Figure S9. Comparison of malignant single-cell expression profiles across gliomas types. (A) Pairwise correlation between the average expression profiles of malignant cells from 25 gliomas, ordered by hierarchical clustering (top) and marked by clinical classification (color bar). **(B-E)** Differentially expressed genes across gliomas. **(B)** Average relative expression of 1,081 genes (rows) determined as differentially expressed across 25 gliomas samples (columns). **(C)** The expression of the genes from (B) in five other IDH-wildtype GBM samples, analyzed in our earlier study (4). **(D)** Genes marked if located on

chromosome 10 (frequently deleted in GBM), arms 1p or 19q (deleted in IDH-O). (E) Enrichment of PRC2 targets (*10*) in each subset of differentially expressed genes (Fisher's exact test).

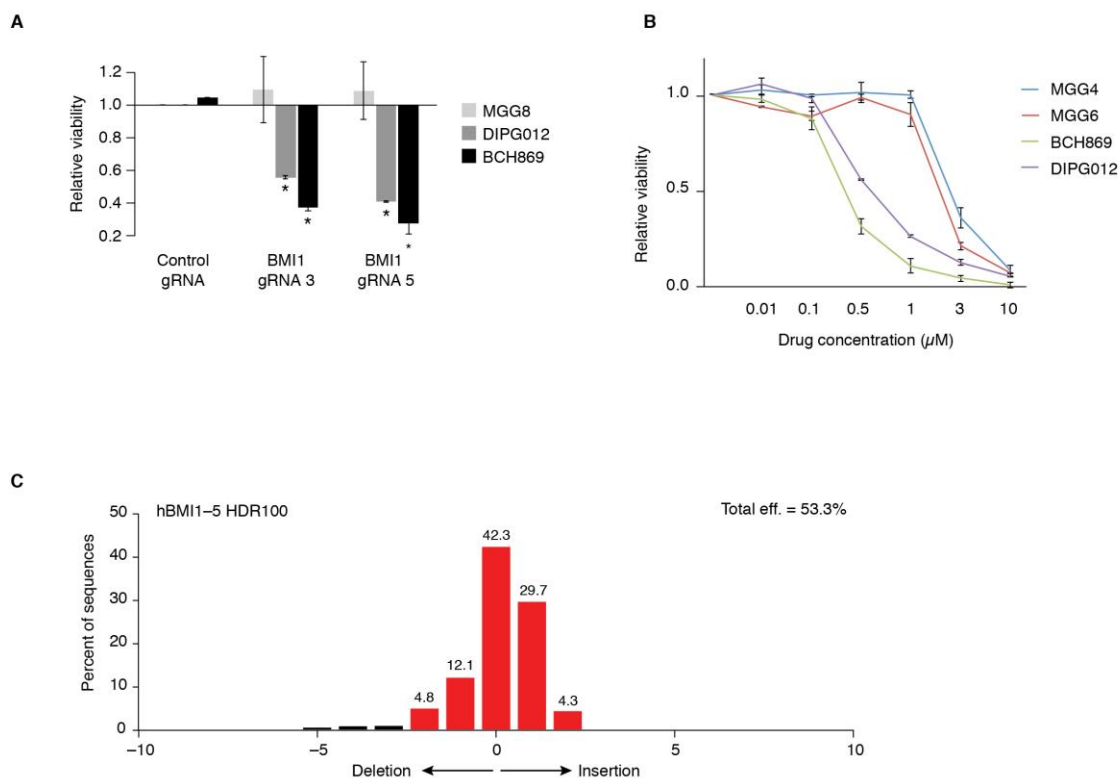


Figure S10. Treatment with BMI-1 inhibitor and BMI-1 CRISPR cutting efficiency. (A) Relative viability (Y axis) of H3K27M (BCH869, DIPG012) and H3 wildtype (MGG8) patient-derived cell lines 7 days after *BMI1* knockout with each of two different single guide RNAs (sgRNA) relative to non-targeting sgRNA (control); $n = 4$ repeats; two technical replicates in two independent experiments. $* = p < 0.05$ by paired, two-tailed t-test. Error bars: $\pm$ S.E.M. **(B)** Relative viability (Y axis) of H3K27M (BCH869, DIPG012) and *H3F3A* wildtype (MGG4, MGG6) patient-derived cell lines following treatment with different concentration (X axis) of the BMI1 inhibitor PTC209. Error bars: SEM. **(C)** Representative cutting efficiency of *BMI1* gene. Distribution of the frequencies of insertions and deletions (Y axis) for gRNA5 in BCH869. Total cutting efficiency is 53.3%.

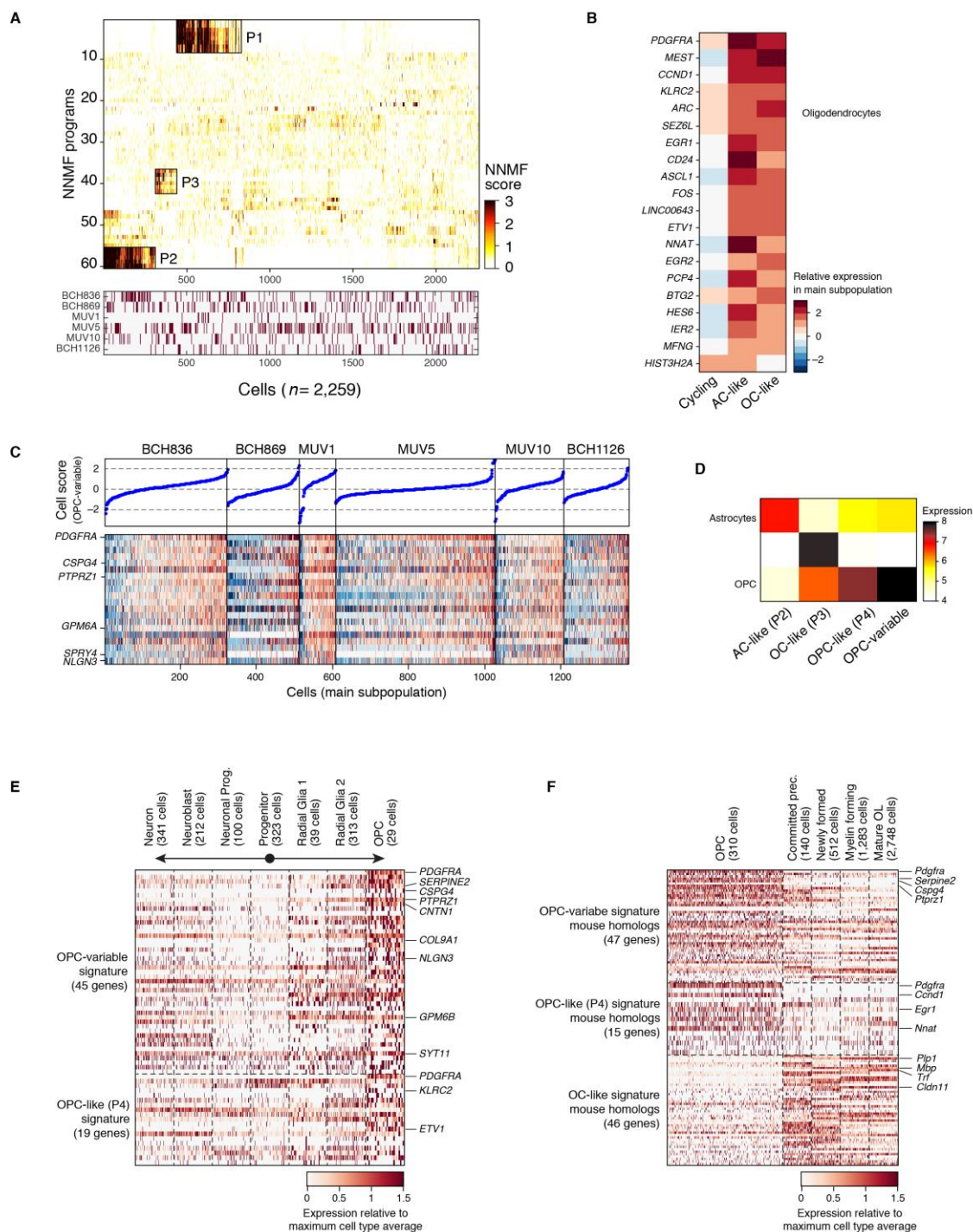


Figure S11. Intra-tumor expression programs in H3K27M-gliomas and their comparison to normal development. (A) Top: scores (color bar) of 2,259 malignant cells (columns, from six H3K27M-glioma tumors) for each of 60 programs (rows) derived from NNMF analysis of individual H3K27M-glioma tumors. Cells and programs are hierarchically clustered, and three sets of correlated programs (P1, P2 and

P3) are highlighted; bottom: assignment of cells (columns, ordered as in top panel) to the six tumors (rows).

(B) Average expression (log-ratio) of P4 (**Fig. 2**) compared to two of the three other subpopulations (OC-like, AC-like and cycling cells; (8)); genes are rank ordered by average upregulation from top to bottom. 19/20 genes are upregulated compared to both AC-like and OC-like cells, but not cycling cells and form the OPC-like program (8).

(C) OPC-variable gene signature, defined by PCA (8). Bottom: Relative expression of the 20 top-scoring genes (rows) of the OPC-variable program across the six tumors; in each tumor, cells are (columns) are ordered by their OPC-variable program scores (shown in top panel).

(D) Average expression in normal brain cell types (rows) of the five programs (columns) identified in analysis of intra-tumor heterogeneity: AC-like (P2), OC-like (P3), OPC-shared (P4) and OPC-variable.

(E) Scaled expression levels (color bar) of the genes (rows) in the OPC-variable (top) and OPC-shared (bottom) gene signatures in each of 1,357 single cells (columns) from the developing human ventral midbrain (embryonal week 6 to 11), grouped by cell type as described in (13).

(F) Expression levels of the genes (rows) in the OPC-variable, OPC-shared, and OC-like signatures in each of 4,993 single cells (columns) from normal oligodendrocyte lineage differentiation in mouse (postnatal day 21 to 60), grouped by cell type as described in (14).

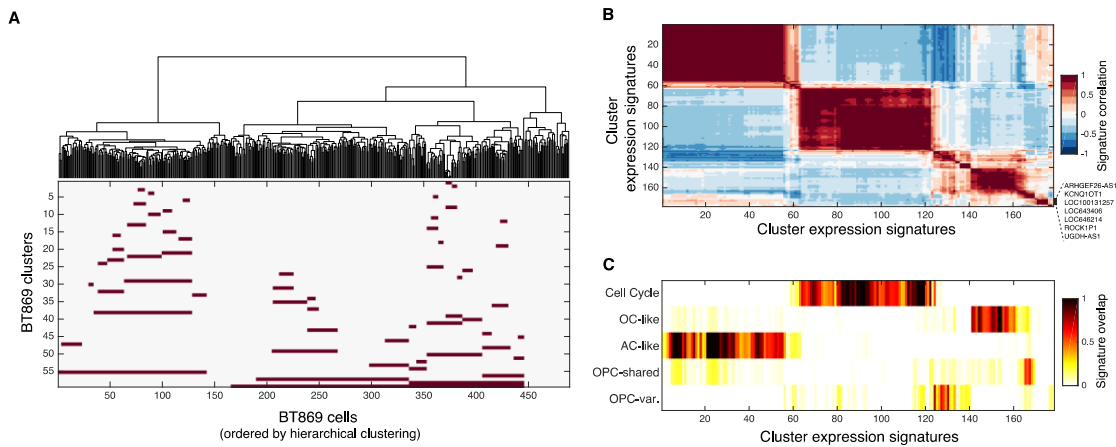


Figure S12. Meta-programs of intra-tumor expression identified by cell cluster signatures or NNMF are highly congruent. (A) Example of cell clusters for signature detection. Malignant cells from BT869 (columns) and the clusters (row) selected by our approach (8) from the full hierarchical tree (top), to which they are assigned (red: cell assigned to cluster). (B) Signature clusters. A gene signature was defined for each selected cluster (8) and scored in each malignant cell, and the signatures were then compared to one another by the correlations in their scores. Shown are the Pearson correlation coefficients between each pair of the 178 signatures (rows, columns), arranged by hierarchical clustering. Clusters were associated with the same functions as the NNMF meta-programs, except for one minor cluster (bottom right corner) that was associated with high expression of pseudogenes, antisense transcripts and uncharacterized transcripts, which might reflect technical rather than biological aspects. (C) Agreement between cluster signatures and NNMF programs. Fraction of genes (color bar) in each of the NNMF meta-programs (rows) that overlap each cluster signature (columns; ordered as in (B)). Apart from the pseudogene-associated cluster, both approaches revealed the same programs.

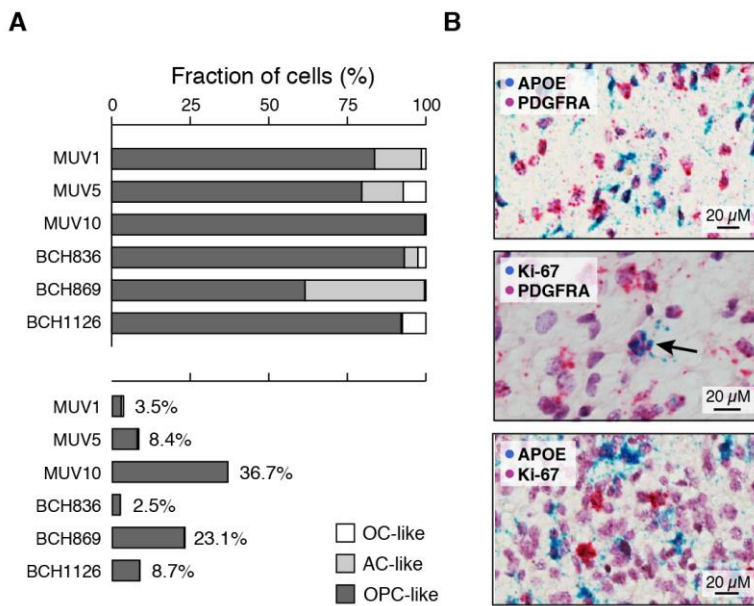


Figure S13. OPC-like cells are the main subpopulation of malignant cells in H3K27M tumors and the source of cycling cells. (A) Proportion of cells (x axis) classified in each tumor as OPC-, AC-, or OC-like across all cells (top), or only among cells classified as cycling (bottom). (B) *In situ* RNA hybridization of H3K27M glioma for astrocytic-like (*APOE*), OPC-like (*PDGFRA*) and proliferation (*Ki-67*) markers; arrows highlight cells co-expressing *PDGFRA* and *Ki-67*.

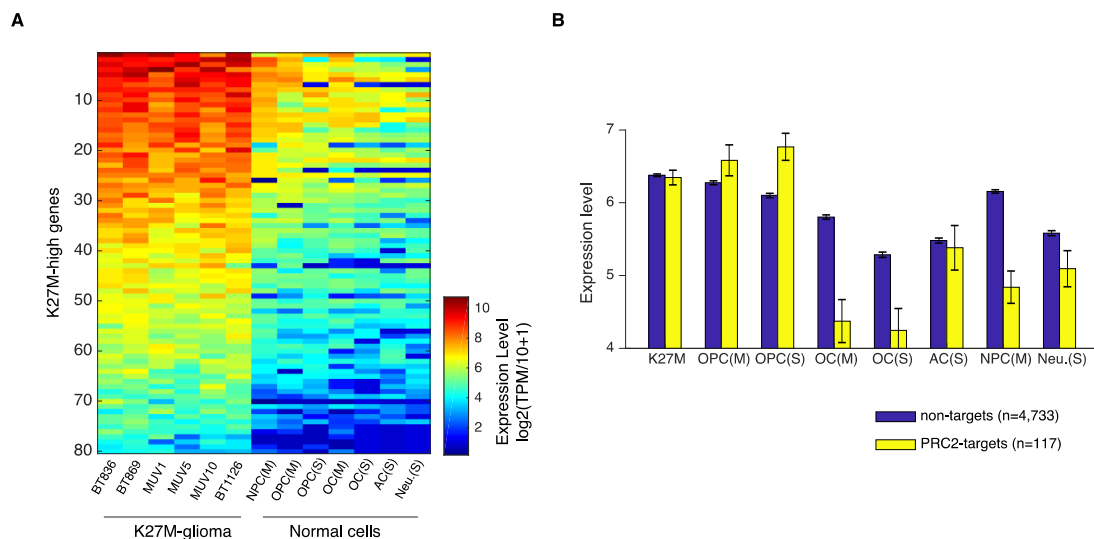


Figure S14. Comparison of gene expression between K27M-glioma and normal brain cell types. (A) Genes with higher expression in H3K27M-glioma malignant cells than in normal brain cell types (8). Shown are the average expression levels (colorbar) of the 80 genes (rows) in the malignant cells of each of the six H3K27M gliomas (left columns), and in seven datasets measured at MGH (“M”) or in a published dataset (11) (“S”) in five cell types: neural progenitor cells (NPC), oligodendrocyte progenitor cells (OPC), oligodendrocytes (OC), astrocytes (AC), and neurons (Neu). Genes are ranked by average expression levels. **(B)** Average expression level (Y axis) of all genes expressed by H3K27M-glioma malignant cells ($E_a > 5$), partitioned into PRC2-targets (yellow) and non-targets (blue) (10), in different sample types (X axis) (8). Error-bars: s.e.m.

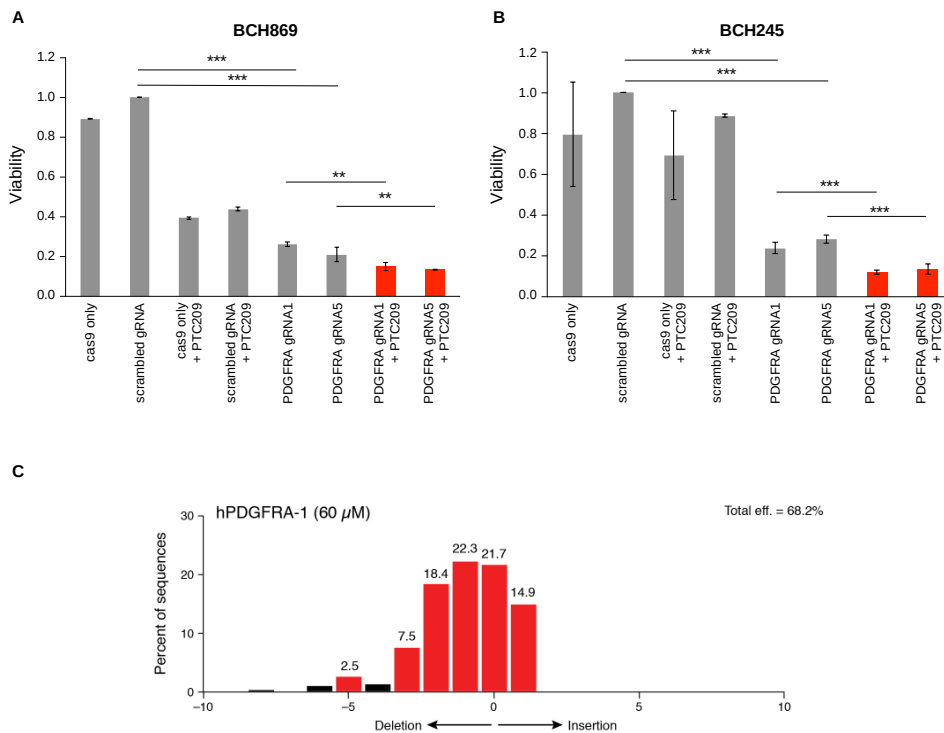


Figure S15. PDGFRA knockout in H3K27M cell lines. (A) Viability of H3K27M patient-derived cell line BCH869 after 7 days of *PDGFRA* knockout with two different gRNAs (1 and 5), normalized to non-targeting gRNA (8), alone or in combination with the BMI1 inhibitor PTC-209 at 500nM. Three technical replicates were performed in each of two independent experiments for a total of n=6. Error bars represent the s.e.m. Significance was analyzed using a mixed effects linear regression model for the normalized viability data, with a random effect for experiment and fixed effects for drugs. $P < 0.01$ (**), $P < 0.001$ (***) (B) Viability of H3K27M patient-derived cell line BCH245 after 7 days of *PDGFRA* knockout with two different gRNAs, normalized to non-targeting gRNA, alone or in combination with the BMI1 inhibitor PTC-209 at 500nM. Three technical replicates were performed in each of two independent experiments for a total of n=6. Error bars represent the s.e.m. Significance was analyzed using a mixed effects linear regression model for the normalized viability data, with a random effect for experiment and fixed effects for drugs. $p < 0.001$ (***) (C) Representative cutting efficiency of *PDGFRA* gene. Frequencies of insertions and deletions for gRNA1 are shown. Total cutting efficiency equals 68.2%.

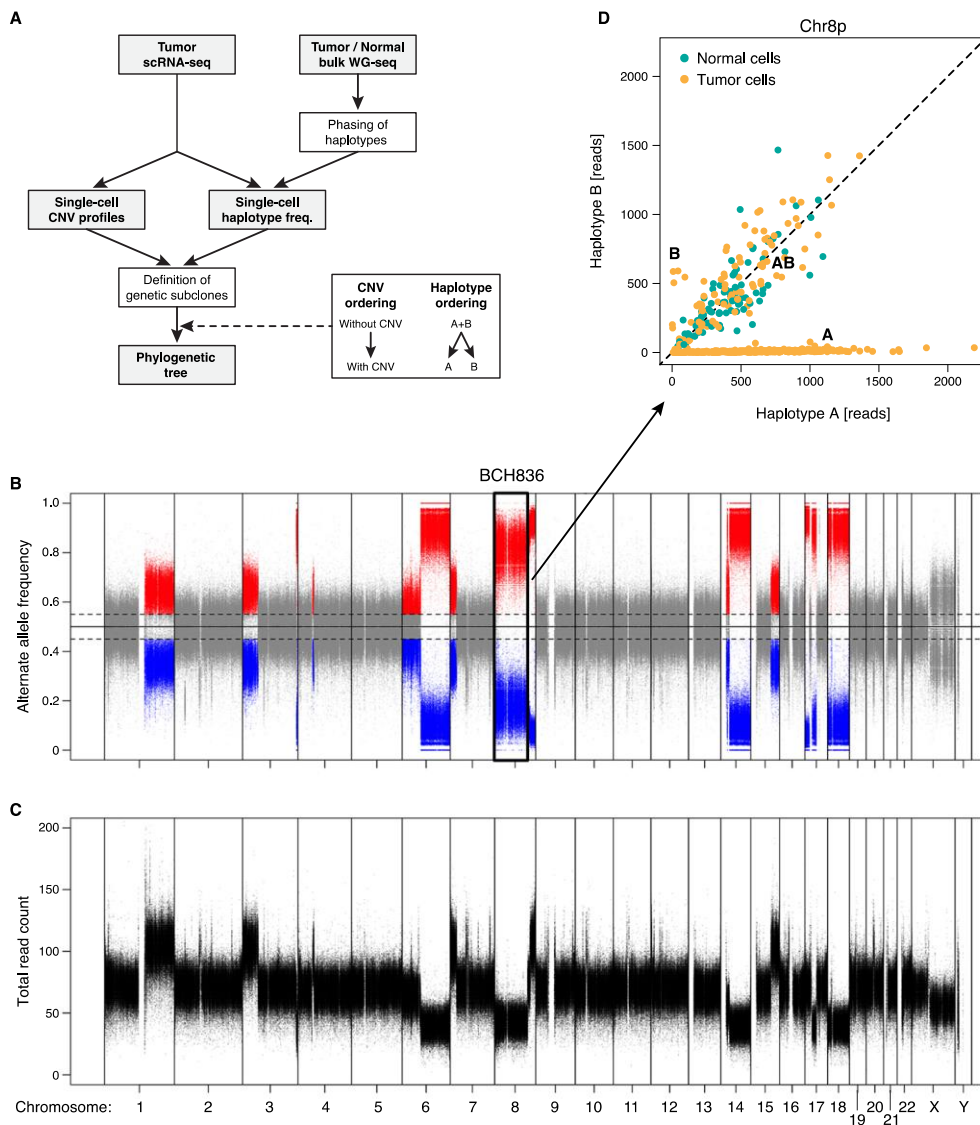


Figure S16. Combined CNV and haplotype analysis from scRNA-seq data. (A) Flowchart illustrating the computational approach: Patient-specific heterozygous SNPs are defined from germline control WGS data (displaying a normal karyotype). (B, C) The allele frequencies of the heterozygous SNPs are then used to phase haplotypes of chromosomes or sub-chromosomal regions that show copy-number variations in the tumor WGS data (e.g. gains and losses, as colored). Rare events, such as copy-number neutral loss-of-heterozygosity of chr17q (harboring *TP53*, which is mutated in this patient), are also detected. (D) The

allele frequency of the combined SNPs of each haplotype are then quantified in the scRNA-seq data. Chromosome 8p is shown as an example, in which most tumor cells score for haplotype A, few tumor cells score for haplotype B, and a number of cells tumor score for both haplotypes, as do the normal cells in this tumor. The combined analysis of single-cell derived CNV profiles and haplotypes frequencies improves the ability to define genetic subclones and helps the inference of their evolutionary relationships.

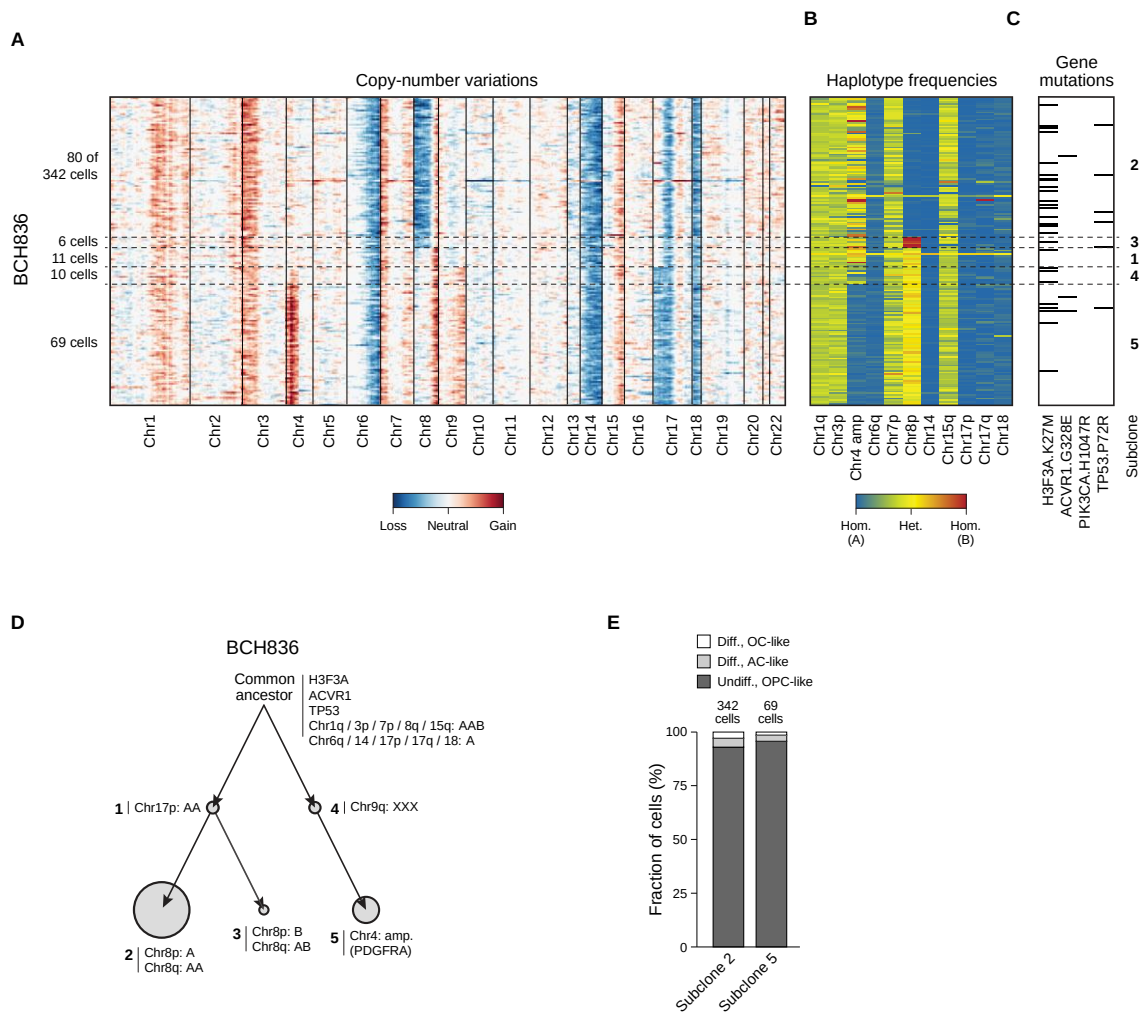


Figure S17. Genetic subclones in H3K27M-gliomas. (A-C) CNVs (A), haplotype frequencies (B) and point mutations in in selected genes identified by WGS (C) as inferred from scRNA-seq for individual malignant cells (rows) from BCH836. Dashed lines: five subclones based on CNV and haplotype profiles (8). 80 out of 342 cells for the predominant subclone 2 are shown. (D) The inferred phylogenetic tree of the individual subclones detected for BCH836 (8). Circle sizes: proportion of cells in the respective subclone. Genetic events (CNVs and gene mutations) are indicated at the inferred point of their first occurrence. (E) Proportion of cells (Y axis) classified into OPC-, AC-, or OC-like states for BCH836 subclone 2 and 5.

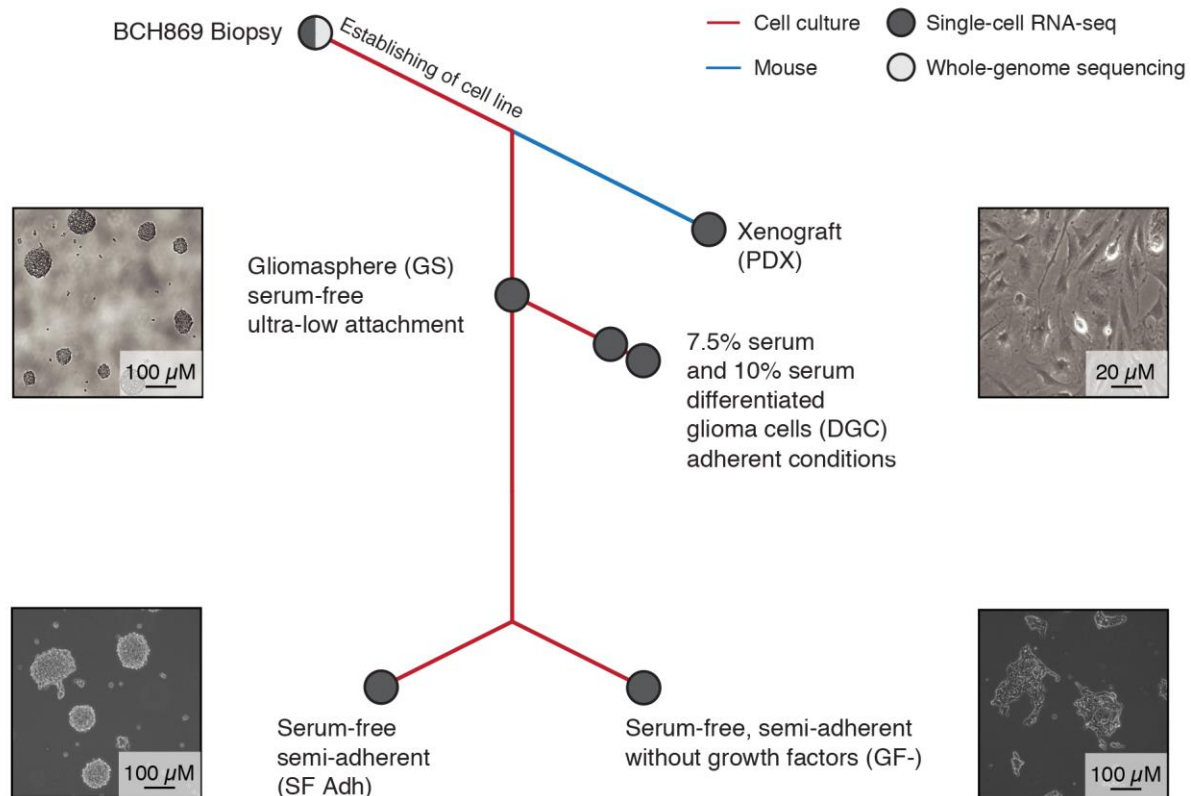


Figure S18. Derivation of BCH869 *in vitro* and *in vivo* models. Schematic showing generation and growth conditions of *in vitro* and *in vivo* models from patient sample BCH869. We expanded BCH869 (i) as gliomaspheres (GS) in serum-free conditions that represent putative tumor-propagating cells; (ii) in serum-free conditions on semi-adherent plastic with or without growth factors (SF_Adh and GF-, respectively); and (iii) in adherent conditions with the addition of serum (7.5% or 10%) as putative differentiated glioma cells (DGC). Representative phase-contrast images are shown for all *in vitro* models.

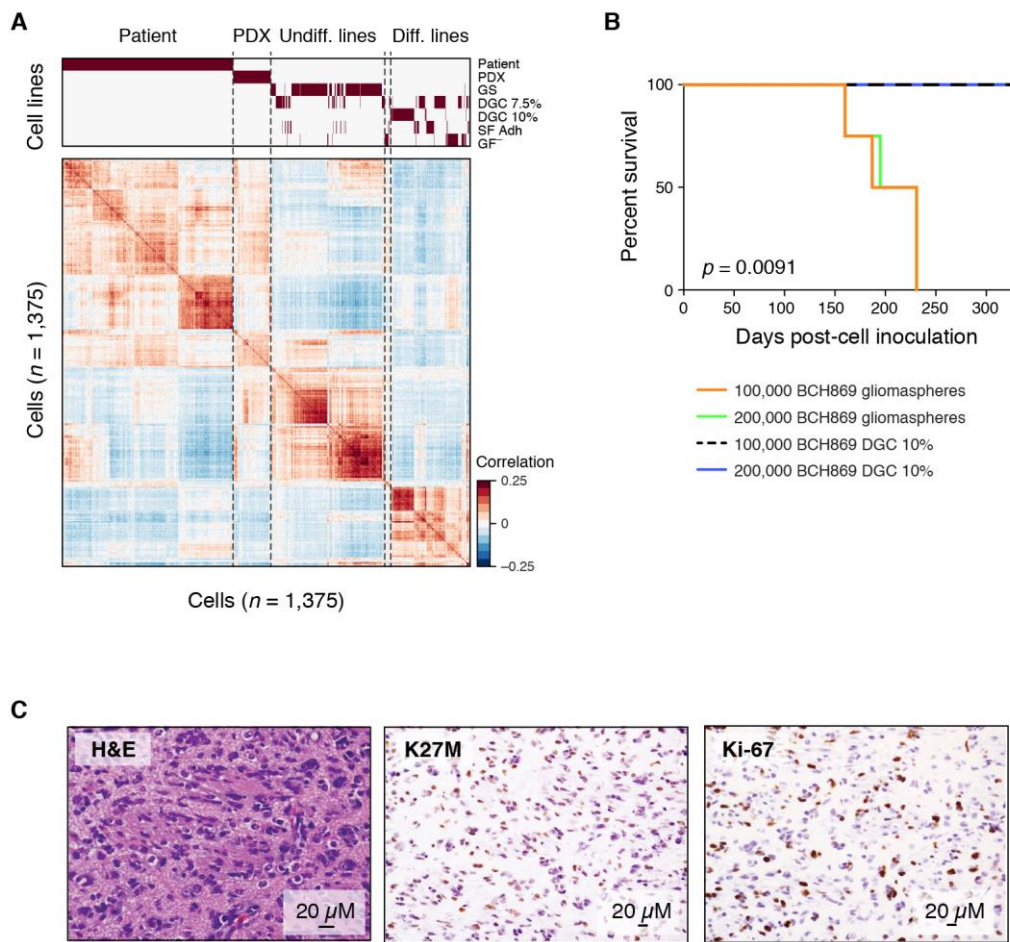


Figure S19. Characterization of BCH869 *in vitro* and *in vivo* models (A) Heatmap depicts pairwise correlations between the expression profiles of cells from tumor, PDX and culture models (8). Cells ordered by sample type (patient, PDX and culture) and within each sample by hierarchical clustering; top panel: assignment of cells to samples; bottom panel: cell-to-cell correlations. The clustering partitioned two groups of cultured cells (“differentiated” and “undifferentiated”). (B) Kaplan-Meier plot of percent of surviving mice (y axis) for mice injected with BCH869 cells grown as gliomaspheres or as DGC 10% (8); $p=0.0091$ Log-rank (Mantel-Cox) for OPC-like vs. differentiated at each cell dose (8). (C) Hematoxyline & Eosin (H&E) stain of BCH869 xenograft showing high cellularity and cytonuclear pleomorphism, H3K27M immunohistochemistry shows nuclear staining of malignant cells. Ki-67 staining highlights proliferative cells.

Supplemental Tables: Table S1 to S8 are hosted online as separate Excel files.

Table S1: Clinical and molecular characteristics of H3K27M-glioma samples profiled by scRNA-seq

Table S2: Single-cell cohort characteristics

Table S3: Differentially expressed genes between glioma types

Table S4: Supplementary patient annotations

Table S5: Gene expression signatures in H3K27M-glioma

Table S6: Differentially expressed genes in H3K27M-gliomas vs normal cell types

Table S7: Subclones based on haplotype and copy number state in BCH836 and BCH869

Table S8: Differentially expressed genes between each pair of sample types

References and Notes

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